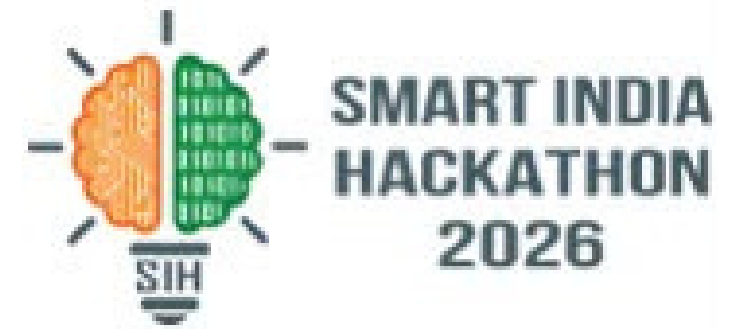


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID** – [SIH26024](#)
- **Problem Statement Title**- [AI-Based Smart Governance and Compliance Monitoring System for Coal Mines](#)
- **Theme**- [Smart Automation](#)
- **PS Category**- [Software](#)
- **Team ID**-
- **Team Name (Registered on portal)**- [DeluluDevs](#)



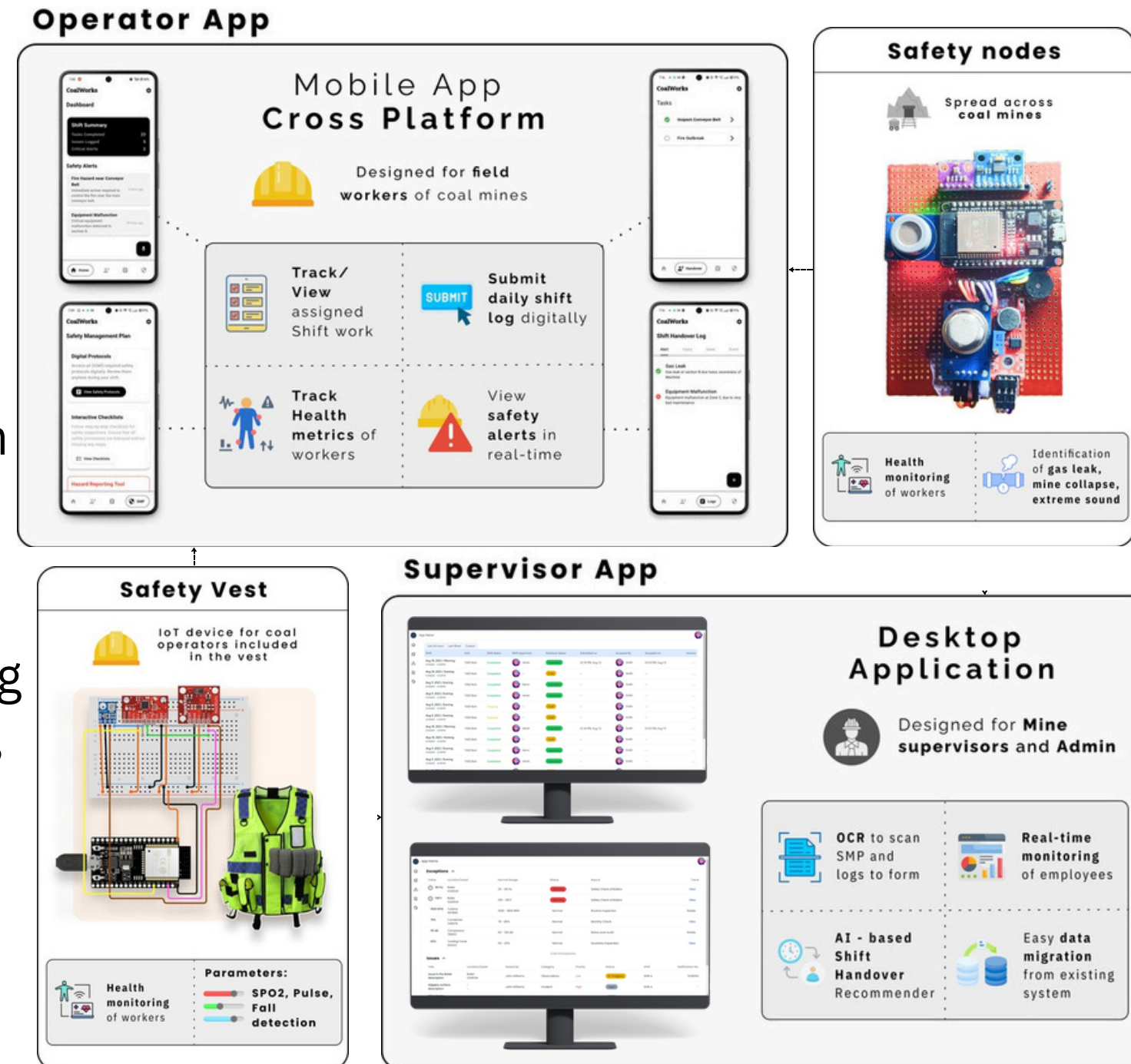
SOLUTION

An AI-based Smart Governance and Compliance Monitoring System for Coal Mines, digitizing Safety, Environment, Production, and Labour regulations into a single automated pipeline.

PROTOTYPE

- **Web dashboard for supervisors** to monitor activities, plan rounds, track tasks, and handover shifts
- **AI-powered assistant** suggests round planning and assigns the right personnel to handle issues
- **OCR** is used to scan and convert paper logbooks into easy template based digital forms
- **Mobile app for workers** to view shift details and log events like issues, alerts, injuries, etc.
- **Automated CAPA Workflows** engine dispatches instant alerts for hazards.
- **Easy data migration** from existing ERP systems using data mapping and APIs

Transforming coal mine management from manual paperwork and fragmented processes into a proactive, automated, data-driven, and closed-loop governance ecosystem for improved efficiency and accountability.

WHY WE STAND OUT ?**Cross-Platform Solution with Docker Integration**

Our Product ensure consistent performance and accessibility across different environments.

**Blockchain Backed Cryptographic Audit**

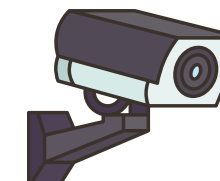
Uses SHA-256 hashing to guarantee tamper proof safety records before government audits.

**Predictive Analytics & Risk Forecasting**

Our AI predicts hazards, plans shifts, assigns tasks, optimizes resources and safety.

**Statutory DGMS Form Auto-Exporter**

Eliminates the need of manual paperwork by auto populating official Forms B, D, and E for regulators.

**Real-Time CV Safety & PPE Scanner**

YOLOv8 computer vision analyzes CCTV feeds for violations like no hardhats or no vests.

BACKEND ARCHITECTURE

Migration from ERP using Data Mapping and APIs

ERP
Existing ERP integrated into our system

LOCAL DB
To store Relational Data

Data Privacy & Security

Zero Trust Ideology

Intranet-only access, blocking external connections

Data Encryption & Hashing

Data is secured through encryption and hashing

Google translation API

API GATEWAY



DOCKER CONTAINER



MongoDB + Beanie
Multi-tenant hierarchical store

Redis Pub/Sub
Real-time alert fan-out & cache

FastAPI Gateway
Async REST API & WebSocket layer

JWT + RBAC
5 roles, Worker to Regulator

Docker Compose
One-command microservice deploy

AI ENGINE - INTEL CORE



Groq LLM (Llama-3)
Free-text issue triage & severity

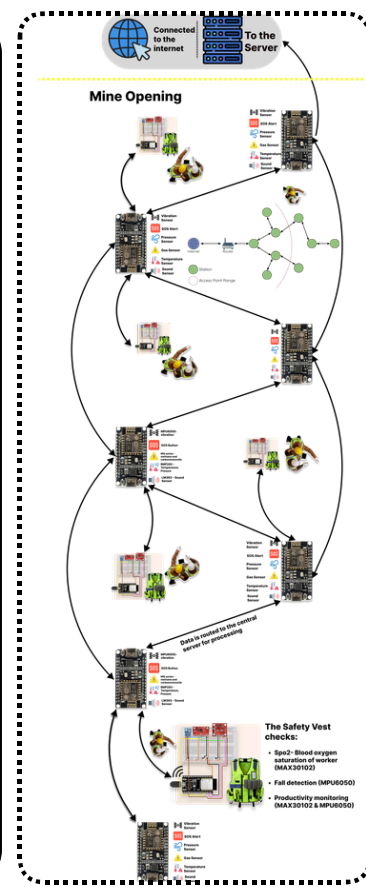
RAG - ChromaDB
Statutory clause retrieval (LangChain)

YOLOv8 CV Service
PPE detection on CCTV frames

DeepFace + MediaPipe
1:1 biometric attendance match

Prophet + IsoForest
EC-cap forecast & anomaly score

FLASK



TECHNICAL APPROACH

ADMIN

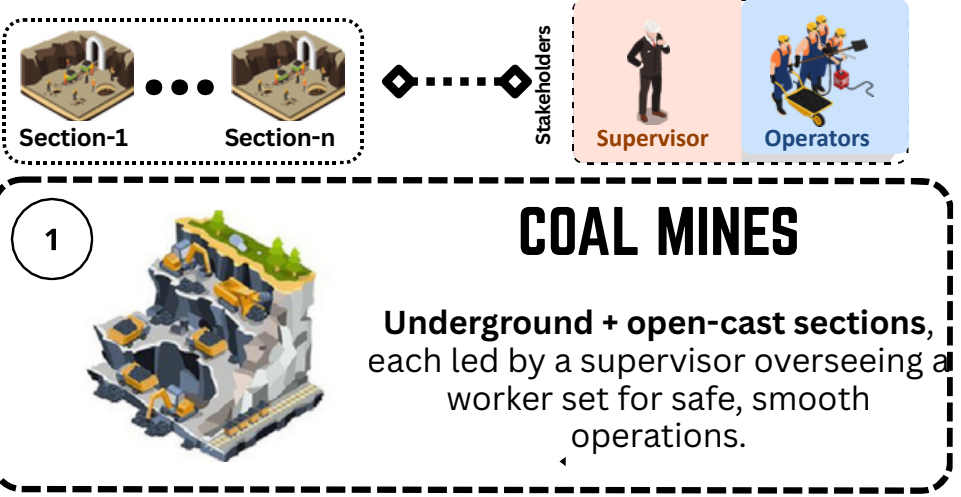
Efficient Task Handling with Web App for Admin

Task Handling
Full worker provisioning & mapping

Zero-Trust Security
Intranet-only, blocks external access

DGMS Report Verify
Admin-signed compliance certificates

Multilingual OCR
Legacy logbook digitization



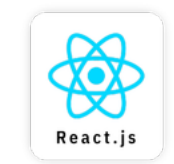
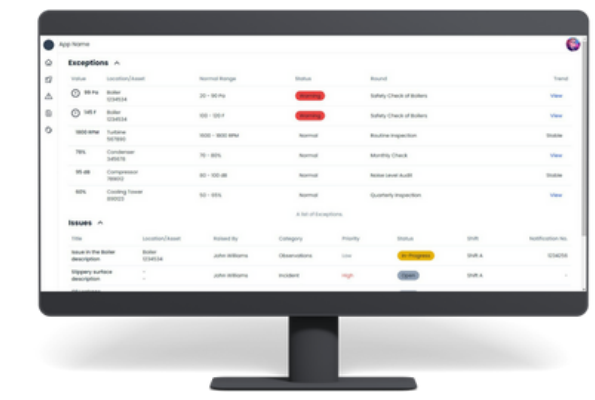
COMPLIANCE & AUDIT ENGINE

- RAG Citation
"Violation of DGMS Rule 124(2)"
- SHA-256 Ledger
Tamper-proof chained audit trail

END-TO-END DATA FLOW

- Field PWA → FastAPI Gateway → MongoDB
- CCTV → YOLOv8 → CAPA Ticket
- IoT Sensors → MQTT → WebSocket → Dashboard
- Free-text → Groq LLM → RAG Engine → CAPA + Legal Citation

SAFETY OFFICER DASHBOARD

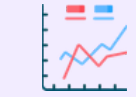


Live GIS Mine Map
Heat-map, hazard pins, EC bounds



Unified Issue Feed
Site + person issues, severity-ranked

WebSocket Alerts
WhatsApp + email via n8n, in-SLA



Real-time IoT logs
Predictive Simulation Engine



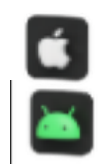
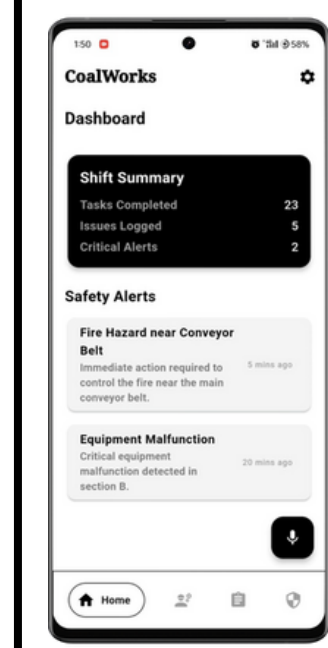
DGMS Compliance
Log Report Submission with Admin Verification

EC Cap Forecasting
Prophet: 30-day breach warning

IOT TELEMETRY & SENSOR LAYER

- ESP-Mesh Fallback
- SpO₂ Vest
- Temp °C
- Air Velocity
- CH₄ / CO ppm

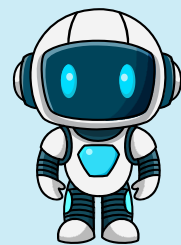
WORKER PWA / MOBILE



- Flutter PWA
- SQLite offline-sync
- Bhashini voice
- DeepFace geo-fence attendance
- Support for major Indian languages

FEASIBILITY & VIABILITY

Feasibility



Technical Feasibility

AI driven risk prediction, compliance monitoring and role-based dashboards integrate with IoT/sensor data and existing mine workflows.



Operational Feasibility

Worker, Mine Safety Officer, Corporate,Regulatory and Admin dashboards centralize alerts, inspections incidents and compliance records.



Economic Feasibility

Built with scalable web technologies (React, Node.js/Express, MongoDB) to reduce infrastructure complexity and support phased deployment.



Regulatory Feasibility

Automates tracking, audit trails, safety reporting and escalation, improving transparency for mine operators and regulators.

Viability



Market Viability

Growing adoption of AI, IoT and digital monitoring in mining creates a strong market opportunity for CoalGuard AI.



Cost Viability

Reduces dependence on manual inspections, paperwork and fragmented monitoring, lowering operational and compliance costs.

Challenges



Integration with Existing Mine Operations

-  **Language & Workforce Adoption**→ Multilingual interface and simple role-based dashboards.
-  **Existing System Integration**→ API-ready architecture for connecting ERP and mine management systems.
-  **Training & Adoption** → Hands-on training and role-specific workflows for workers and officers.



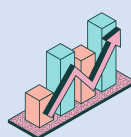
Reliable Safety Intelligence

-  **Sensor/Data Quality** → Data validation and anomaly detection to improve reliability.
-  **False Alerts** → AI cross-validation using multiple data sources before generating critical alerts.
-  **Scalability** → Modular IoT/edge architecture supports additional sensors and mine sites.



Connectivity, Security & Privacy

-  **Poor Internet Connectivity**→ Local/offline data buffering with synchronization when connectivity returns.
-  **Data Security**→ Role-based access, authentication and end-to-end encryption.
-  **Privacy & Access Control**→ Zero-trust approach ensures users only access authorized information.



Scalability

The platform can expand from a single mine to multiple sites, with centralized corporate and regulatory monitoring.



Sustainable Viability

Continuous safety monitoring, predictive insights and digital compliance support safer and more sustainable mining operations.

IMPACT AND BENEFITS

Social & Safety Benefits

Enhances worker safety and accountability through early risk detection, timely alerts, faster response, and transparent corrective-action tracking.

Economic Benefits

Improves operational efficiency by reducing downtime, minimizing manual monitoring efforts, optimizing resources, and lowering incident-related losses.

Environmental Benefits

Promotes sustainable mining through continuous compliance monitoring, digital environmental records, and faster response to potential environmental risks.

Stakeholders & Impacts



Notifies an exposed conveyor hazard and reports it by voice through the mobile app

FIELD WORKERS



Voice log auto-captures geo + ISO timestamp



Faster hazard reporting, less paperwork



Reduced risk exposure from real-time logging



The RAG engine retrieves the applicable DGMS/CPCB clause and assigns a priority ticket

MINE SAFETY OFFICER



Ticket status: assigned, SLA timer started



Zero-hallucination legal citations



Automated SLA escalation, nothing missed



Cross validation of work using data and our AI agent

MINE SUPERVISOR



Approves task completion for smoother handover



Simplified audits with real-time tracking



Continuously safety tracking prevents potential risks



Compliance dashboard updates instantly with production, safety and risk metrics

MINE MANAGEMENT



Utilizes data analytics and compliance



Data-driven decisions across every mine



Early warning before compliance caps are breached



Government & Regulatory Bodies

- Real-time data improves oversight and compliance monitoring

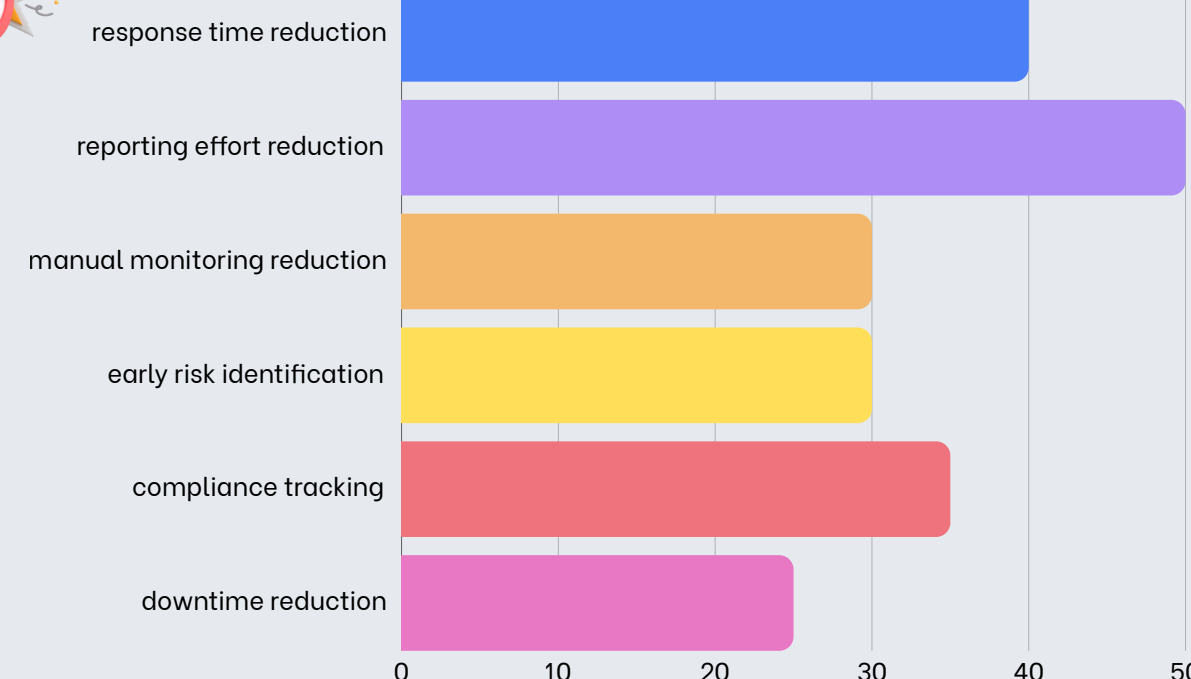
- Immutable audit trail supports targeted, evidence-based regulation

TRANSFORMING MINE SAFETY OPERATIONS

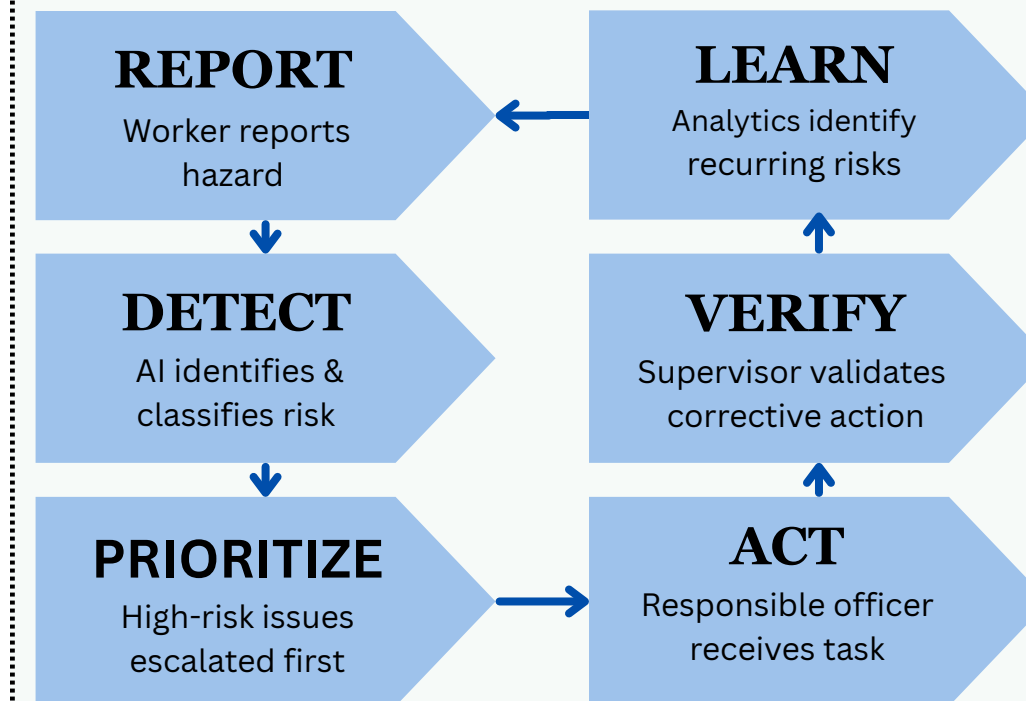
TODAY — Reactive	COALGUARD AI — Proactive
Manual hazard reporting	Digital + voice-assisted reportig
Scattered records	Centralized safety database
Delayed escalation	AI-prioritized alerts
Manual compliance tracking	Automated compliance monitoring
Paper-heavy audits	Evidence-backed digital audit trail
Reactive decisions	Data + AI-assisted decisions
Difficult trend identification	Continuous risk analytics



Expected Impact



IMPACT CREATION LOOP



RESEARCH AND REFERENCES

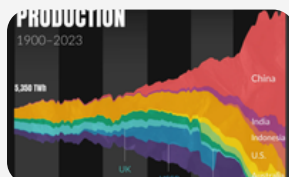
COAL MINES IN INDIA



India's coal mining began in 1774, evolving into vast subsidiaries with increasingly complex governance.

Production Stats

Crores of tonnes mined annually across 100s of sites make manual compliance tracking unsustainable.



Coal as a Vital Energy Source

Coal fuels India's steel and power sectors, making compliant mining a national priority.

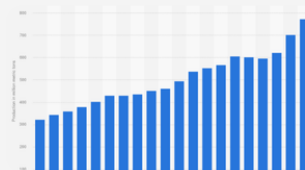


Production Stats

India's production in global coal stats

India's Global Coal Position

As the world's 2nd-largest coal producer, coal sector needs governance that scales across 100s of mines



POWER UTILITY COAL DESPATCH

Rising dispatch volumes mean more field activity, inspections, and compliance data across every mine.

DGMS REGULATORY AND WORKING

DGMS regulates coal mines safety in India, enforcing standards under the Mines Act to protect workers and machinery. It also conducts inspections, accident investigations, and safety audits to ensure compliance.

They publish reports and case studies, which are valuable references for researchers analyzing mine safety trends, accidents, and technological advancements.

COMPLIANCE SOFTWARE MARKET

Compliance Governance Software Market

India's GRC software market was valued at ~\$1,788M in 2025, projected to reach \$4,443M by 2034.



Per-Mine SaaS Deployment

Number of Workers: 500
Compliance Dashboard Seats: 20 units
Mobile Field-App Licenses: 50 units

Dashboard Licenses

Units: 20
Price per Unit: ₹8,000/yr
Total: ₹1,60,000

Mobile Field-App Licenses

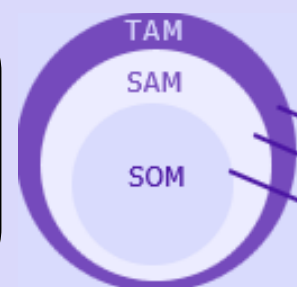
Units: 50
Price per Unit: ₹4,000/yr
Total: ₹2,00,000

Installation Fees

Units: 1 (per mine)
Price: ₹1,50,000
Total: ₹1,50,000

Profit Percentage

Software Licensing: Profit margins on SaaS licensing (85%) due to low marginal cost of scaling across mines. Implementation & Support: Profit margins on onboarding/integration (35%) due to per-site setup effort.



TAM - Total Addressable Market
SAM - Service Available Market
SOM - Service Obtainable Market

Total Revenue

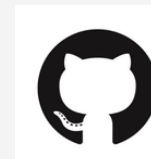
Year-1: Pilot, 1 mine
₹6,90,000

Service and audits
AI compliance & anomaly-detection module: Monthly Cost: ₹15,000"
Total: ₹1,80,000/yr

[CLICK HERE](#)

OUR WORK

GITHUB



You can see the project CoalGuard by clicking here

WEBSITE



Click here to see the website

GIS MAPPING



Leaflet.js + React-Leaflet

single-window GIS-based environment/forest/wildlife clearance system



OpenStreetMap Tile API

Open-source base maps visualizing coal mines and field activity across India.

BACKEND



FASTAPI

Async Python web framework powering the backend REST API and AI engine microservice.



MongoDB

NoSQL document database serving the primary data store, accessed through Motor and Beanie ODM.

AI ENGINE

PPE Detection
YOLOv8 + OpenCV



Compliance RAG Engine
LangChain + Groq + ChromaDB



BLOCKCHAIN

Solidity + OpenZeppelin

Immutable AuditLedger smart contract ensuring tamper-proof compliance records with secure access control.



Hardhat + Ethereum Sepolia

Smart contract development, testing, and on-chain report hash anchoring.



FRONTEND

React + Vite + Tailwind

Modern component-based UI enabling fast builds and responsive dashboards for field officers and admins.



Zustand + ECharts

Lightweight state management for real-time compliance dashboards.

